॥ श्रीगणेशाय नमः ॥

Interest

Simple interest

$P = ₹10000 \quad R = 10\% \quad N = 2$ yrs

1st year → $10\% \cdot 10000 = ₹1000$
2nd year → $10\% \cdot 10000 = ₹1000$
3rd year → $10\% \cdot 10000 = ₹1000$

$A = P + I = 10000 + 3000 = ₹13,000$

Compound interest

$P = 10000 \quad n = 3 \quad R = 10\%$ p.a.

1st year → $10\% \cdot 10000 = ₹1000$
2nd year → $10\%(10000) + 10\%(1000) = 10\%(11000) = ₹1100$
3rd year → $10\%(10000) + 10\%(1000) + 10\%(1100) = 10\%(12100) = ₹1210$

$A = 10,000 + (1000 + 1100 + 1210) = ₹13,300$

Simple interest

$$SI = \frac{Pnr}{100}$$

Simple interest is calculated according to the following formula:

$$\frac{r \cdot B \cdot m}{n}$$

where

r is the simple annual interest rate

B is the initial balance

m is the number of time periods elapsed and

n is the frequency of applying interest.

Q. $P = ₹10000$, $r = 5\%$ p.a., $n = 3$ yrs. SI.

→ $A = P + SI = 10000 + \frac{10000 \cdot 3 \cdot 5}{100} = ₹11,500$

Q. $r = 5\%$ quarterly. $n = 36$ months. $P = ₹10000$. SI

→ $r = 5 \times 4 = 20\%$ p.a. $\quad n = 3$ yrs

$SI = \frac{10000 \times 3 \times 20}{100} = 6000 \quad A = ₹16000$

Q. Dinesh deposited rs. 65,800. SI at 14% p.a. for 4 yrs. Amount he'll get?

→ $A = 65800 + \frac{65800 \times 4 \times 14}{100} = 65,800 + 36,848 = ₹1,02,648.$

Q. Mr.A takes a loan from Mr.B at 5% p.a. and returns ₹6,600 after 2 years to complete the payment. What's the interest paid?

→ $n = 2$ yrs, $r = 5\%$ p.a

$$A = P + \frac{Pnr}{1000} = P\left(1 + \frac{nr}{100}\right)$$

$$6600 = P\left(1 + \frac{10}{100}\right) = P \cdot \frac{11}{10}$$

$P = 6,000$

Interest $= 6,600 - 6,000 = ₹600$

Q. SI for a sum of rs. 1230 for 2 years is rs.10 more than SI on rs. 1130. Rate?

→ $\frac{1230 \cdot 2 \cdot r}{100} - \frac{1130 \cdot 2 \cdot r}{100} = 10$

$\frac{100 \cdot 2 \cdot r}{100} = 10$

$r = 5\%$ p.a.

Q. SI for a sum for 2 years at 8% p.a. is rs. 110 more than that for 1 year at 5% p.a. for the same sum. Find the sum.

→ $\frac{P \cdot 8 \cdot 2}{100} - \frac{P \cdot 1 \cdot 5}{100} = 110$

$11P = 11{,}000$

$P = ₹1{,}000$

Q. Mr.R lent rs.400 to A for 2 years and rs.100 to M for 4 years. He got rs. 60 as interest from both. Find the rate of interest.

→ $\frac{400 \cdot 2 \cdot r}{100} = 8r$

$\frac{100 \cdot 4 \cdot r}{100} = 4r$

$12r = 60$

$r = 5\%$ p.a.

Q. Arjun borrowed rs.800 at 6% and Rajesh borrowed rs.600 and 10%. After how much time will they both have equal debts?

→ $800\left(1 + \frac{6 \cdot n}{100}\right) = 600\left(1 + \frac{10n}{100}\right)$

$4\left(\frac{50+3n}{50}\right) = 3\left(\frac{10+n}{10}\right)$

$200 + 12n = 150 + 15n$

$50 = 3n$

$n = 16.66$ yrs.

Compound interest

$A = P + I$

$A = P\left(1 + \frac{R}{100}\right)^n$ ← duration

The total accumulated value, including the principal sum P plus compounded interest I, is given by the formula:[8] [9]

$$A = P\left(1 + \frac{r}{n}\right)^{tn}$$

where:

- A is the final amount
- P is the original principal sum
- r is the nominal annual interest rate
- n is the compounding frequency
- t is the overall length of time the interest is applied (expressed using the same time units as r, usually years).

Q. CI for rs.8000 after 3 years at 5% p.a. ?

→ 1st year → $8000 \cdot \frac{5}{100} = 400$

2nd year → $8400 \cdot \frac{5}{100} = 420$

3rd year → $(8000 + 400 + 420) \cdot \frac{5}{100} = 441$

$CI = 400 + 420 + 441 = ₹1{,}261$

$A = ₹9{,}261$

using formula → $A = 8000 \times \left(1 + 5\%\right)^3$

$= 8000\left(\frac{105}{100}\right)^3 = 8000 \times \frac{21}{20} \cdot \frac{21}{20} \cdot \frac{21}{20} = 21^3 = ₹9261$

$CI = A - P = ₹1260$

Q. CI for rs.30,000 at 7% p.a. for a certain time is rs.4347. What's the duration?

→ $A = 30000 + 4347 = 34347 = P\left(1 + 7\%\right)^t = 30000\left(\frac{107}{100}\right)^t$

$\left(\frac{107}{100}\right)^t = \frac{34347}{30000} = \frac{11449}{10000} = \frac{107^2}{100^2}$

$t = 2$

Q. If a sum of rs.1000 amounts to rs. 27,000 in 3 years, what's the rate of CI p.a.?

→ $27000 = 1000\left(1+\frac{r}{100}\right)^3$

$\left(1+\frac{r}{100}\right)^3 = 27 = 3^3$

$r = 200\%$

$A = P\left(1+\frac{r}{n}\right)^{tn}$ (t: time, n: compounding period)

Compounded half-yearly

Q. A invested rs. 16,000 at the rate of 10% p.a. for 1 year. Interest is compounded half-yearly. Find the total amount received by A at eoy.

→ $A = 16000\left(1+\frac{10\%}{2}\right)^{2\times 1} = 16000\left(\frac{105}{100}\right)^2 = 1.6\times(105)^2 = 17640$

$\frac{12}{6}$ ← year, ← ½ year

OR

$P = 16000 \quad r = 5\% \quad t = 2$ (1 year has 2 6 month periods)

$A = 16000\left(1+5\%\right)^2 = ₹17,640.$

Q. A sum of rs. 40,000 is invested for 18 months at 20% p.a. compounded half-yearly. What's the interest to be paid?

→ $A = (40,000)\left(1+\frac{20\%}{2}\right)^{\frac{18}{12}\times 2} = 40000\left(\frac{110}{100}\right)^3 = ₹53,240$

$CI = ₹13,240$

Q. Find the CI on rs. 32,000 for 6 months at 10% p.a., the interest being compounded quarterly.

→ $A = 32,000\left(1+\frac{10\%}{4}\right)^{\frac{6}{12}\times 4} = 32,000\left(\frac{102.5}{100}\right)^2 = 32,000\times(1.025)^2 = 33,620$

$\frac{12}{3}$

$CI = 33,620 - 32,000 = ₹1,620$

Q. A sum of rs. 10,000 was deposited in a bank for a period of 27 months at the rate of 20%.

→ 27 months = 24+3 = 2yrs 3months

1st year → $\frac{20}{100}\times 10000 = 2000$

2nd year → $\frac{20}{100}\times 12000 = 2400$

3 months → $\frac{(20/4)}{100}\times 14400 = 720$

CI = 5120

A = 15120

Using formula: (2yrs, 3 months)

$A = 10,000\left(1+20\%\right)^2\left(1+\frac{20\%}{4}\right)^{4\times\frac{3}{12}}$

$= 10000\left(\frac{120}{100}\right)^2\left(\frac{105}{100}\right)^1 = ₹15,120$

Q. P=10,000. R=12% p.a. T=27 months.

→ $10,000\left(1+12\%\right)^2\left(1+\frac{12\%}{4}\right)^1$

Q. P=10,000. R=12% p.a. T=29 months.

→ $10,000\left(1+12\%\right)^2\left(1+\frac{12\%}{(12/5)}\right)^1 = 10000\left(\frac{112}{100}\right)^2\left(\frac{105}{100}\right)^1 = 13171.2$

Q. P=10,000. R=12% p.a. T=33 months. Compounded quarterly.

→ 33 months = 11 quarters

Rate = 12% p.a = 3% per quarter.

$$A = 10,000\left(1 + 3\%\right)^{11}$$

Q. What will be the total amount received for rs.20,000 for a period of 30 months with 8% p.a.

→ $A = 20000\left(1 + 8\%\right)^{2}\left(1 + \frac{8\%}{2}\right)^{1} = 24261.12$

Population increase/decrease

Say we have an increase of 10% every year.

2017	2018	2019	2020
1000	1010	1210	1331

Q. The population of a city increases by 15% p.a. If it was 4000 at the end of year 2015, what'll it be at the end of 2017?

→ $4000\,(1+15\%)^2 = 5290$

Q. The population of a city decreases by 10% p.a. If it was 4,00,000 at the end of year 2017, what'll it be after 3 years?

→ $4,00,000\,(1-10\%)^3 = 2,91,600$ (rate is negative)

Q. The population of a city increases by 5% p.a. If the present population is 3,70,440, what was it 3 years ago?

→ $3,70,440 = x\,(1+5\%)^3$

$x = 3,20,000$

Q. The population of a city is 5000. It increases by 10% in the first year, decreases by 20% in the 2nd year, increases by 30% in the third. What'll be the population after all of this?

→ $\left(\left(\left(5000 \times \frac{110}{100}\right) \times \frac{80}{100}\right) \times \frac{130}{100}\right) = 5720$

Q. A sum of money amounts to rs. 4840 in 2 years, and to rs. 5324 in 3 years with compound interest compounded anually. Rate of interest?

→ $4840 = P\left(1+\frac{r}{100}\right)^2$

$5324 = P\left(1+\frac{r}{100}\right)^3$

$\Rightarrow \frac{5324}{4840} = \left(1+\frac{r}{100}\right)^2$

$(1.1) = \left(1+\frac{r}{100}\right)$

$\frac{r}{100} = 0.1$

$r = 0.1 \times 100 = 10$ $\quad r = 10\%$ p.a.

Q. A sum of money amounts to rs. 800 in 3 years, and to rs. 882 in 5 years with compound interest compounded anually. Rate of interest?

→ $\frac{882}{800} = \frac{P\left(1+\frac{r}{100}\right)^5}{P\left(1+\frac{r}{100}\right)^3}$

$\frac{441}{400} = \left(1+\frac{r}{100}\right)^2$

$1+\frac{r}{100} = \frac{21}{20}$

$\frac{r}{100} = \frac{1}{20}$

$r = 5\%$ p.a.

Q. A sum of money amounts to rs. 7,000 in 4 years, and to rs. 10,000 in 8 years with compound interest compounded anually.
The initial amount of money invested is?

→ $\frac{10000}{7000} = \frac{P\left(1+\frac{r}{100}\right)^8}{P\left(1+\frac{r}{100}\right)^4}$

$\left(1+\frac{r}{100}\right)^4 = \frac{10}{7}$

$7000 = P\left(1+\frac{r}{100}\right)^4$

$P = 7000 \cdot \frac{7}{10} = 4900$